Model Diagnostics: Computational Practice

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October 29, 2026

Standalone companion to the complete regression diagnostics reference. Personal study notes prepared with AI assistance; not an official course handout or organizational SOP.

1 Real-data R laboratory: industrial stack loss

Question: How does mean stack loss vary with air flow, cooling-water temperature and acid concentration? The built-in `stackloss` dataset contains 21 observations from a nitric-acid plant. The response is an operational measure of ammonia loss; consult the dataset help for its scale and measurement details. This exercise estimates conditional associations, not causal effects. Allow about 45–60 minutes for the full lab, or 15 minutes for plots and influence only.

```
# Reproducible diagnostic lab; run in R. Base R sections need no packages.
# Optional: install.packages(c("lmtest", "sandwich"))
data(stackloss)
help("stackloss")
d <- stackloss
fit <- lm(stack.loss ~ Air.Flow + Water.Temp + Acid.Conc., data=d)
print(summary(fit))
# Standard plot.lm diagnostics: mean, distribution, scale, influence.
pdf("stackloss_diagnostics.pdf", width=9, height=7)
par(mfrow=c(2,2)); plot(fit, which=c(1,2,3,5)); dev.off()
n <- nobs(fit); p <- length(coef(fit))
checks <- data.frame(case=seq_len(n), residual=resid(fit),
  leverage=hatvalues(fit), external=rstudent(fit), cook=cooks.distance(fit))
checks$outlier_screen <- abs(checks$external) > qt(1-.05/(2*n), n-p-1)
print(checks[order(-checks$cook), ])
i <- which.max(checks$cook)
fit_del <- update(fit, data=d[-i, , drop=FALSE])
print(cbind(full=coef(fit), without_case=coef(fit_del)))
print(confint(fit)); print(confint(fit_del))
# Specified curvature: use identical rows in both fits.
fit_curve <- update(fit, . ~ . + I(Air.Flow^2))
print(anova(fit, fit_curve)) # exact classical F requires its assumptions
if (requireNamespace("lmtest", quietly=TRUE)) {
  print(lmtest::bptest(fit)) # studentized Koenker default
  print(lmtest::resettest(fit, power=2:3, type="fitted"))
  # bgtest only if row order has a justified temporal interpretation.
}
if (requireNamespace("lmtest", quietly=TRUE) &&
    requireNamespace("sandwich", quietly=TRUE)) {
  print(lmtest::coeftest(fit, vcov.=sandwich::vcovHC(fit, type="HC3")))
```

```
  # HC3 is not a cure for mean misspecification, dependence or confounding.
}
# Fixed-X normal-model calibration of residual Shapiro-Wilk statistic.
set.seed(761); B <- 999
X <- model.matrix(fit); h <- hatvalues(fit)
stopifnot(all(h < 1))
stat <- function(y) {
  e <- lm.fit(X, y)$residuals
  unname(shapiro.test(e/sqrt(1-h))$statistic)
}
Wobs <- stat(model.response(model.frame(fit)))
Wsim <- replicate(B, stat(rnorm(n)))
p_mc <- (1 + sum(Wsim <= Wobs))/(B + 1)
print(c(W_observed=Wobs, calibrated_p=p_mc,
       approximate_MC_SE=sqrt(p_mc*(1-p_mc)/(B+1))))
# Pointwise Q-Q simulation envelope for internally studentized residuals.
# This is NOT a simultaneous 95% band or a global 5% test.
std_res <- function(y) {
  e <- lm.fit(X,y)$residuals
  sort(e/(sqrt(sum(e^2)/(n-p))*sqrt(1-h)))
}
sims <- replicate(B, std_res(rnorm(n)))
band <- apply(sims,1,quantile,probs=c(.025,.975))
z <- qnorm(ppoints(n)); obs <- sort(rstandard(fit))
pdf("stackloss_qq_envelope.pdf")
plot(z,obs,ylim=range(obs,band),xlab="Normal quantile",
     ylab="Ordered internally studentized residual",main="Pointwise reference envelope")
polygon(c(z,rev(z)),c(band[1,],rev(band[2,])),
       col=adjustcolor("skyblue",alpha.f=.35),border=NA)
points(z,obs,pch=19); lines(z,apply(sims,1,median)); dev.off()
# Under this fixed-X Gaussian null, these standardized residual statistics
# are invariant to beta and positive sigma, allowing rnorm(n) simulation.
# Real prediction exercise: fit lm(mpg ~ wt + hp, mtcars), use leave-one-out
# predictions y_i - e_i/(1-h_ii), and compute RMSE. Repeat all model selection
# inside each training fold if you choose predictors or transformations.
print(sessionInfo())
```

2 Research computation: stability, covariance and reproducibility

2.1 QR and SVD instead of explicitly inverting normal equations

The inverse formula is useful for theory. In computation use a stable least-squares solver. For $X = QR$, with $Q^TQ = I_p$, solve $R\widehat{\beta} = Q^Ty$. Then $H = QQ^T$ and $h_{ii} = \sum_{k=1}^{p} Q_{ik}^2$; a full $n \times n$ matrix is unnecessary for leverage. For rank checks and near collinearity, inspect singular values $X = U\Sigma V^T$. For full rank, $\kappa_2(X) = s_{\max}/s_{\min}$ and $\kappa_2(X^TX) = \kappa_2(X)^2$. Scaling helps conditioning but cannot create information absent from the design. A small normal-equation residual alone cannot certify accurate coefficients in an ill-conditioned problem.

2.2 Coefficient deletion identity

For case design vector x_i (a column) and full-rank deleted design, the Sherman–Morrison identity gives

$$\widehat{\beta} - \widehat{\beta}_{(i)} = \frac{(X^T X)^{-1} x_i e_i}{1 - h_{ii}}.$$

Multiplying by X and using $\|X(X^T X)^{-1} x_i\|^2 = h_{ii}$ produces Cook's formula. This derivation separates mathematical identity from a practical influence threshold.

2.3 HC3 and generalized least squares

For independent observations with potentially unequal variances, an HC3 covariance estimator is

$$\widehat{V}_{HC3} = (X^T X)^{-1} \left\{ \sum_i x_i x_i^T \frac{e_i^2}{(1 - h_{ii})^2} \right\} (X^T X)^{-1}.$$

Its usual inferential justification is asymptotic; it is not an exact small-sample replacement for all classical tests. Cluster dependence needs cluster-aware methods and sufficient independent clusters. For $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 \Omega$ with known positive-definite Ω, factor $\Omega = LL^T$ and whiten: $y^* = L^{-1} y$, $X^* = L^{-1} X$. OLS on the transformed data gives

$$\widehat{\beta}_{GLS} = (X^T \Omega^{-1} X)^{-1} X^T \Omega^{-1} y.$$

Use triangular solves rather than explicit inverses. Diagnostics should examine the transformed error structure and the scientific plausibility of Ω. Estimating Ω introduces additional uncertainty; feasible GLS does not inherit every exact known-covariance result.

2.4 Independent computational checks

The accompanying `verify_diagnostics.py` uses NumPy least-squares/QR and explicit case deletion to verify projection idempotence, residual orthogonality, $\sum h_{ii} = p$, Cook's formula and deleted residual identities. On the seeded synthetic example, maximum absolute discrepancies were 5.41×10^{-15} for Cook's distance and 2.94×10^{-15} for deleted residuals. These validate numerical identities on this example, not universal scientific adequacy. The R laboratory is supplied as reproducible source but was not executed in the authoring environment; run it in R and retain session information before reporting results. A stronger software exercise uses NIST certified reference data across multiple difficulty levels. Compare matched quantities under the exact specified model, intercept convention and units. Agreement checks numerical computation; it does not establish the model's scientific validity.

3 Expert resources and a reading pathway

This is a curated, broad pathway, not an exhaustive bibliography. Links were checked through source discovery on October 29, 2026. Package behavior can change; inspect installed help and session information.

1. **Penn State STAT 501**: Lesson 11: Influential Points; Lesson 4: Model Assumptions. Start here for worked distinctions among outliers, leverage and influence.
2. **NIST/SEMATECH**: Regression diagnostics reference. Use for a broad catalogue of diagnostic ideas and applied checks.
3. **R Core documentation**: plot.lm; influence.measures. Read for exact definitions used by software.
4. **lmtest authors' documentation**: bptest; bgtest; resettest. Study hypotheses, defaults and reference distributions before using tests.
5. **Frank E. Harrell Jr.**: Regression Modeling Strategies course notes. Continue here for specification, nonlinear effects, missing data, overfitting and validation, especially in biomedical work.
6. **R datasets documentation**: Brownlee's Stack Loss Plant Data. Read the primary software dataset record before interpreting the lab.

Suggested sequence: definitions and figures; Penn State examples; execute the lab; consult R/lmtest references; then Harrell's broader modeling strategy. A defensible report connects the scientific target, evidence of departures, and sensitivity of conclusions.